Evolution, 59(11), 2005, pp. 2343–2352

PLEIOTROPIC EFFECTS OF BENEFICIAL MUTATIONS IN *ESCHERICHIA COLI*

ELIZABETH A. OSTROWSKI,[1,2] DANIEL E. ROZEN,[3,4] AND RICHARD E. LENSKI[1,5]
[1]*Ecology, Evolutionary Biology, and Behavior Program, Michigan State University, East Lansing, Michigan 48824*
[2]*E-mail: ostrow24@msu.edu*
[3]*Department of Biology, Emory University, Atlanta, Georgia 30322*
[4]*E-mail: drozen@emory.edu*
[5]*E-mail: lenski@msu.edu*

Abstract.—Micromutational models of adaptation have placed considerable weight on antagonistic pleiotropy as a mechanism that prevents mutations of large effect from achieving fixation. However, there are few empirical studies of the distribution of pleiotropic effects, and no studies that have examined this distribution for a large number of adaptive mutations. Here we examine the form and extent of pleiotropy associated with beneficial mutations in *Escherichia coli*. To do so, we used a collection of independently evolved genotypes, each of which contains a beneficial mutation that confers increased fitness in a glucose-limited environment. To determine the pleiotropic effects of these mutations, we examined the fitnesses of the mutants in five novel resource environments. Our results show that the majority of mutations had significant fitness effects in alternative resources, such that pleiotropy was common. The predominant form of this pleiotropy was positive—that is, most mutations that conferred increased fitness in glucose also conferred increased fitness in novel resources. We did detect some deleterious pleiotropic effects, but they were primarily limited to one of the five resources, and within this resource, to only a subset of mutants. Although pleiotropic effects were generally positive, fitness levels were lower and more variable on resources that differed most in their mechanisms of uptake and catabolism from that of glucose. Positive pleiotropic effects were strongly correlated in magnitude with their direct effects, but no such correlation was found among mutants with deleterious pleiotropic effects. Whereas previous studies of populations evolved on glucose for longer periods of time showed consistent declines on some of the resources used here, our results suggest that deleterious pleiotropic effects were limited to only a subset of the beneficial mutations available.

Key words.—Adaptation, beneficial mutations, correlated response, micromutational models, niche breadth, pleiotropy, trade-offs.

Received March 1, 2005. Accepted August 18, 2005.

The form and extent of pleiotropy is central to many theories in evolutionary biology, including the evolution of specialization (Futuyma and Moreno 1988; Jaenike 1990; Cooper and Lenski 2000), senescence (Rose 1991), and limits to adaptation (Barton and Keightley 2002; Otto 2004). The importance of pleiotropy was emphasized by Fisher, who formalized its role in his geometric model of adaptation (Fisher 1958). This model has two main assumptions: first, that pleiotropy is common, such that most mutations affect many or all of the traits under selection; and second, that the rate of environmental change is slow, such that organisms evolve toward a single, slow-moving optimum. As a consequence of the high ''dimensionality'' associated with adaptation, Fisher reasoned that only mutations of small effect would be adaptive.

Fisher's basic idea that pleiotropy is a common property of mutations has spurred other models that address the kinds of variation that are most likely to contribute to adaptation. For example, Lande (1983) formulated a model in which adaptation can be driven by numerous mutations of small effect or a few mutations of large effect. An important assumption of this model is that large mutations generally have large deleterious pleiotropic effects, whereas small mutations have smaller or no such effects. Unfortunately, evidence to support this assumption is lacking. As Orr and Coyne (1992, p.731) emphasize, not only does it require that highly beneficial mutations have deleterious pleiotropic effects, but the magnitude of these effects must generally outweigh the magnitude of the selected benefit:

> Lande's micromutational theory requires that large mutations be *disproportionately* worse than smaller mutations. We simply have no information here. We do not know, for example, whether mutations adding four bristles to a fly are more than four times as harmful as mutations adding only a single bristle. Moreover, characteristics of deleterious mutations may tell us little about favorable ones. Even if large mutations are less fit than small ones, it does not follow that large mutations with a favorable primary effect suffer more deleterious effects than mutations with smaller primary effects.

In this study, we aim to test these suppositions raised by Orr and Coyne by examining the form and extent of pleiotropy associated with beneficial mutations in *Escherichia coli*. These mutations have a favorable primary effect: they increase fitness in a glucose-limited environment. Using these mutants, we examine the consequences of these mutations for fitness in novel resources. Specifically, we address the following pair of questions: (1) Do mutations that improve fitness in one resource environment have pleiotropic effects on fitness in alternative resource environments? In particular, will improvements in one environment usually entail becoming worse in others? (2) Do pleiotropic effects correlate in magnitude with the advantage in the primary selected environment? For example, will larger improvements entail larger trade-offs?

Several aspects of *E. coli* biology make it a good system for studying adaptation. Bacteria are easily cultured in the laboratory and can be stored in a nongrowing state (a −80°C freezer), enabling direct comparison of evolved and ancestral strains. Short generation times and large population sizes can be achieved in the laboratory, such that adaptation occurs on observable time scales. Finally, replicate populations can be started from a single isogenic ancestor, so that multiple random samples of evolution from a single starting point can be obtained. This aspect allows us to collect a library of different, spontaneous beneficial mutations and to quantify their fitness effects relative to the ancestor.

Previous work has suggested that there is extensive pleiotropy associated with resource use in this system. For example, Remold and Lenski (2001) examined the fitness effects of random insertion mutations in novel resource and temperature environments and found that many of these mutations exhibited differential fitness effects across resource environments. Travisano et al. (1995) found that some populations evolved for 2000 generations in glucose had improved fitness in maltose, but others had reduced fitness. The proposed explanation for this variability was that the populations had fixed beneficial mutations that differed in their pleiotropic effects on fitness in maltose. On other resources, populations showed either consistently increased fitness (N-acetylglucosamine [NAG] and mannitol), or consistently decreased fitness (galactose and melibiose), suggesting that pleiotropy may be common, but that it differs in sign depending upon the resource in question (Travisano and Lenski 1996). Finally, Cooper and Lenski (2000) examined the consequences of 20,000 generations of adaptation to a glucose-limited environment for the ability to catabolize alternative resources. Replicate populations often showed parallel reductions of function, suggesting that the pleiotropic effects of beneficial mutations led to narrower catabolic niche breadth.

These studies have supported a general role for pleiotropy with regard to resource use, but two elements of these studies make it difficult to infer how common pleiotropy is for individual beneficial mutations. In the first study, collection of the mutations was random with respect to fitness (most were, in fact, deleterious) and they were produced by transposon insertions, meaning that they were disproportionately likely to be knock-out mutations; it is not clear how representative this kind of variation is with respect to adaptation. And whereas the long-term evolution experiments suggest that antagonistic pleiotropy underlies losses of function, it is not clear whether these findings were the result of many weakly pleiotropic mutations, or whether they were instead driven by one or a few mutations with highly pleiotropic effects. In general, there is little empirical data on the distribution of pleiotropic effects associated with beneficial mutations, despite the importance of this information to evolutionary theory (Wingreen et al. 2003; Otto 2004). Thus, in the current study, we seek a more direct assessment of the distribution of pleiotropic effects by examining individual beneficial mutations.

Materials and Methods

We used a collection of 30 genotypes that were independently evolved from a single ancestral genotype in an environment where glucose was the limiting carbon source (Rozen et al. 2002). These mutants were collected in such a way as to ensure that, in all likelihood, they each contain exactly one beneficial mutation. For each of these 30 mutants, we determined its fitness relative to the ancestor in glucose (the ''selective'' environment) and in five other resources (the ''novel'' environments).

Choice and Significance of Novel Resources

The novel resources we use vary in the extent to which their uptake and catabolism differs from that of glucose (Travisano and Lenski 1996). Two of the novel resources, NAG and mannitol, use the phosphoenolpyruvate transferase system (PTS) for transport across the inner membrane of the *E. coli* cell, which is also used for glucose transport. Resources that are transported using this system are phosphorlyated upon entry into the cell (Postma et al. 1996). The non-resource-specific enzymes HPr and EI pass the phosphoryl group to the various resource-specific (EII) transporter proteins. By contrast, the three other novel resources we use—maltose, galactose, and melibiose—are non-PTS resources and use resource-specific permeases for transport into the cell. PTS enzymes are known to inhibit directly the uptake and metabolism of non-PTS resources, resulting in the preferential use of glucose even in the presence of other resources. In studies of populations evolved for 2,000 generations in a glucose-limited medium, these resources showed a mixture of correlated responses ranging from primarily positive to primarily negative (Travisano and Lenski 1996).

Bacterial Strains and Culture Conditions

All genotypes used in this study were derived from the same *E. coli* B ancestor (Lenski et al. 1991). This ancestor contains no plasmids and is strictly asexual. In this system, different genotypes can be distinguished by the Ara marker, which denotes the ability (Ara$^+$) or inability (Ara$^-$) to catabolize L(+)arabinose. Genotypes with the Ara$^-$ or Ara$^+$ marker state form red or white colonies, respectively, when plated on tetrazolium arabinose indicator agar. This color difference allows us to track the relative proportion of two genotypes (with opposite marker states) when we plate mixed cultures on indicator agar during competition assays.

In the current study, all mutants evolved from an ancestral strain that carried the Ara$^+$ marker. When assessing relative fitness, the evolved Ara$^+$ genotypes were competed in mixed cultures against the Ara$^-$ variant of the ancestral strain. Previous studies have shown that the Ara$^+$ and Ara$^-$ variants of the ancestor are neutral in the same glucose-limited culture conditions that were used in the current study (Lenski et al. 1991; Travisano et al. 1995). Moreover, we performed our own control competitions between the Ara$^-$ and Ara$^+$ ancestral variants in glucose, as well as in the five novel resources. The results confirm that this marker difference is neutral in all resources used in the experiment (data not shown). Although this requirement is not strictly necessary, it means that competitions between Ara$^+$-evolved genotypes and the Ara$^-$ variant of their ancestor can be assumed to be equivalent to competitions between the evolved genotypes and their Ara$^+$ (true) ancestor, with the marker serving only to allow the competitors to be distinguished.

Collection of Mutants

Collection of the mutants was described in detail in Rozen et al. (2002). Briefly, 30 replicate populations were founded with both Ara^+ and Ara^- variants of the ancestor in the following ratios (Ara^+:Ara^-): 1:100, 1:10, 1:1, 10:1, and 100:1. The populations were propagated daily by transferring 0.1 ml of the previous day's culture to 9.9 ml of fresh glucose medium. This 100-fold dilution permits a 100-fold daily increase in population size, equivalent to approximately 6.6 (or $\log_2 100$) generations per day. While the populations were being propagated, they were also plated periodically to assess the relative proportions of the two marker types. A sustained deviation in the ratio of the two marker types indicated the rise of a beneficial mutation in the increasing subpopulation, at which point transfers were ceased for that population and clones of both marker types were isolated and saved. No population evolved beyond 400 generations, regardless of whether such a deviation had been observed.

Samples to determine relative frequencies of both markers were taken every day for the first seven days and then every other day thereafter. Monitoring of the trajectories at such high resolution allowed rapid and reliable detection of deflections from initial ratios. In cases where deflections occurred prior to 400 generations, clones from earlier time points—near the point of deflection—were used. The maximum duration of the experiment was limited to 400 generations to minimize the possibility of fixation of double mutations. This time scale was chosen based upon earlier work (Lenski et al. 1991; Elena et al. 1996; Gerrish and Lenski 1998) that provided estimates for beneficial mutation rates and expected times to fixation for populations evolving under identical conditions.

For the purposes of the current study, we independently verified that a beneficial mutation was indeed present in each of these clones (see Statistical Analyses). In addition, it is highly probable that each of these mutants contains only a single beneficial mutation, owing to the discrete shifts in relative abundance used to detect the mutations and the short duration of the experiment. A convenient aspect of our bacterial system is that reproduction is asexual and thus multiple beneficial mutations must arise sequentially on a single genetic background. Isolation of the mutants immediately after a shift in relative abundance in the mixed population makes it unlikely that a second beneficial mutation would have had sufficient time to arise on the same genetic background. To verify this expectation, Rozen et al. (2002) used a maximum likelihood approach to estimate the rate of beneficial mutations, using empirical estimates of the selection coefficients of the mutations and the deviation in marker ratios at the time of collection. Using these estimates, they found that in 100 simulations of their experiment, none of the ''winning'' clones (those whose frequency was increasing at the time of collection) had obtained more than one mutation.

Fitness Assays

Each mutant's relative fitness was determined via head-to-head competition assays with its ancestor in glucose (the selective environment) and five novel resources. The competition assays were performed as described by Lenski et al. (1991). Briefly, both competitors were inoculated from the freezer into 9.9 ml of Luria broth and allowed to grow overnight. Next, as a preconditioning step, each mutant was transferred (via a 10,000-fold dilution) into each of the media types to be tested—generally glucose and one of the novel resources—and allowed to grow and acclimate for an additional 24 h at 37°C. In all cases, Davis minimal medium was used, supplemented with 25 μg/ml of whichever carbon source was being tested.

The competition began when an equal volume (0.05 ml) of each competitor's pre-conditioned culture was transferred into a single test tube containing 9.9 ml of fresh medium and the appropriate carbon source. The competitors were then allowed to grow together for 24 h, during which time they competed for the same pool of limiting resource. The competition cultures were plated at the beginning and end of the 24-h period to assess the initial and final densities of the two competitors. From these densities, we calculated each competitor's realized Malthusian parameter, that is, its net rate of population growth during the competition. Relative fitness was then calculated simply as the ratio of the realized Malthusian parameters of the two competitors (Lenski et al. 1991).

We ran the competition experiments as sets of complete blocks. Within each block, we competed all 30 mutants against the ancestor in glucose and in one other resource. We then replicated this complete-block design three times for each of the five novel resources. Using this design, we obtained three independent fitness estimates for each mutant in the five novel resources, for a total of 450 competitions (30 genotypes $\times$ 5 resources $\times$ 3 blocks). In addition, fitness in glucose was measured in every block, such that we had a total of 450 estimates of fitness in glucose (30 genotypes $\times$ 15 blocks).

Of these 900 estimates of relative fitness, experimental errors led to the loss of two values: one in glucose and one in galactose. One additional missing value was introduced when we discarded a fitness estimate in glucose that was an extreme statistical outlier. This estimate was more than eight standard deviations above the mean estimate in glucose. The loss of these estimates resulted in a very slight imbalance in our datasets; *F*-tests must therefore be considered approximate.

Statistical Analyses

Inclusion of mutants in the study

To establish that all of the presumptive mutants did, in fact, carry beneficial mutations, we calculated a mean and standard error for the fitness of each mutant in the selective (glucose) environment. As a preliminary step, we used a conservative two-tailed *t*-test to ask whether a mutant's relative fitness differed significantly from a null-hypothesized value of one. If we failed to detect a statistically significant difference, we then performed a series of additional six-day competitions against the reciprocally marked ancestor in glucose. These longer competitions permit resolution of very small differences in fitness, enabling us to distinguish between a mutation of small effect and the absence of a mutation. Those few mutants that could not be confirmed to have increased fitness even following these more sensitive tests were dropped from subsequent analyses.

Heterogeneity of fitness effects in the selective environment

To determine the heterogeneity of fitness effects in the selective environment, we performed an analysis of variance (ANOVA) based on fitness estimates of the mutants in glucose only. This analysis was performed in SAS using PROC GLM (SAS Institute 1999), with block and genotype coded as random effects. Because each block contained a single replicate of each genotype, a genotype × block interaction was not included in the model. To quantify the heterogeneity in fitness effects in glucose due to differences among genotypes, we calculated a variance component for the effect of genotype. This component was calculated as the difference between the mean squares for genotype and error, divided by the sample size.

Pleiotropic effects of mutations in novel resources

To determine whether a particular mutant had pleiotropic effects in novel resources, we used two-tailed *t*-tests to assess whether its relative fitness in each resource differed from a null-hypothesized value of one, which would indicate fitness equal to the ancestor. We then assessed the heterogeneity of pleiotropic effects as a function of genotype and resource by performing a two-way ANOVA. This ANOVA was calculated using PROC GLM in SAS, with genotype and resource treated as random and fixed effects, respectively. *F*-tests were constructed according to Scheffé (1959), where the random effect of genotype was tested over the mean square error, and the effect of resource was tested over the interaction of genotype × resource. We also ran five separate one-way ANOVAs to assess the effect of genotype separately for each novel resource (Appendix Table A1). Similarly, we ran separate one-way ANOVAs to examine the effect of resource for each genotype (Appendix Table A2). These additional analyses aided the interpretation of the main effects of genotype and resource.

Scaling of pleiotropic effects

To address whether pleiotropic effects were proportional to the direct effects, we calculated Pearson correlation coefficients between fitness in glucose and each of the five novel resources. Prior to the analysis, we assessed normality of the data using a Shapiro-Wilkes test in SAS. The correlation coefficients and their *P*-values were then obtained using SYSTAT (2002).

Our pairwise experimental design, in which genotypes competed against the ancestor simultaneously in glucose and in one other resource, permitted two possible routes for calculating correlation coefficients. When comparing fitness effects of each mutant in glucose and another resource, we could limit our glucose estimate to the mean of the three estimates that were measured concurrently with the three estimates in the novel resource. This method would use a mean fitness in glucose based on fewer replicates, but controlled for possible variability across blocks. Alternatively, we could use the fitness in glucose calculated as the average of all 15 estimates. This method improves the accuracy of the estimate, but potentially conflates differences across blocks. After performing the analysis both ways, we found that, in all cases, the two methods produced the same significant results. For clarity, we present the analysis where a genotype's fitness in glucose is based on all 15 estimates, which makes it easier to compare the fitness effects of mutants across resources. However, because we used the same glucose estimates in all five correlations, we applied a sequential Bonferroni correction (Rice 1989) to these correlation results.

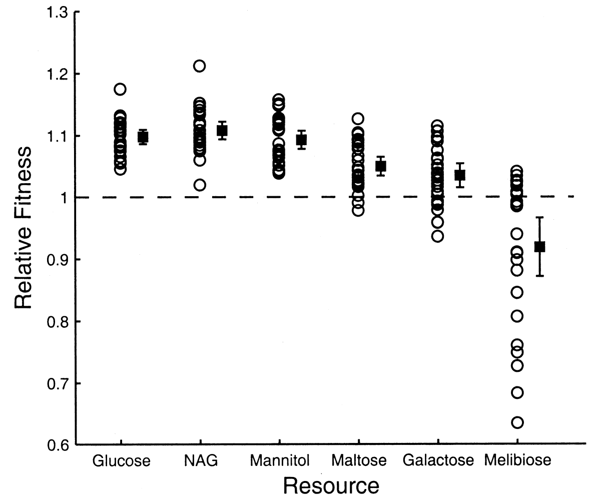

FIG. 1. Relative fitness of the 27 mutants in glucose and the five novel resources. Each circle represents the mean fitness of a mutant based on three independent measures (except in glucose, where each value is the mean of 15 independent measures). Squares represent the grand mean fitness in each resource, and error bars represent 95% confidence intervals based on $n = 27$.

Where there was a significant correlation between fitness in glucose and a novel resource, we also calculated the slope of a reduced major axis (Model II) regression, according to the following formula (Sokal and Rohlf 1995): $b_{RMA} = b_{OLS}/r_{XY}$, where b_{OLS} is the slope of the ordinary least squares regression, and r_{XY} is the correlation coefficient between the fitness values of mutants in glucose and the alternative resource. The correlation coefficients, slopes of the least squares regressions, and their standard errors were calculated in SYSTAT. Using these estimates, we calculated confidence intervals on b_{RMA} as described in Sokal and Rohlf (1995).

RESULTS

Size and Heterogeneity of Fitness Effects in the Selective Environment

Examination of the fitness values of the mutants in the selective glucose environment revealed that 27 of the 30 mutants tested were significantly more fit than the ancestor. The results of the additional six-day competitions confirmed that the remaining three mutants did not differ from the ancestor in fitness (data not shown). These three mutants were dropped from further analyses, owing to the lack of evidence that they carried a beneficial mutation.

The remaining 27 mutants had a mean fitness in glucose of 1.096 (Fig. 1, far left). Results of the analysis of variance

TABLE 1. Analysis of variance for fitness effects in glucose. Values are based on 15 independent estimates for each genotype, with three missing values in total. Block and genotype are random effects.

Source	df	MS	F	P
Genotype	26	0.01273	5.48	<0.0001
Block	14	0.01245	5.36	<0.0001
Error	360	0.00232		

also revealed a significant effect of genotype (Table 1, $P < 0.0001$), indicating that these mutants are heterogeneous in the selective environment. There is also a significant block effect (Table 1, $P < 0.0001$), which indicates some fitness variation due to uncontrolled temporal differences in the environment (e.g., incubator temperature) across replicates. However, our pairwise experimental design, in which each genotype was simultaneously competed against the ancestor in glucose and one other resource in each block of the experiment, allows us to control for this variation across blocks. Finally, the variance component for genotype (calculated as the difference in the mean squares for genotype and error, divided by the sample size) is equal to 0.0007, and the square root of this estimate is 0.0264. In other words, these mutants have an average fitness effect of approximately 10% in glucose, and a typical mutant differs from that average by roughly 3%.

Do Beneficial Mutations Have Pleiotropic Effects in Other Resources?

Next, we sought to determine whether mutations that are beneficial in glucose have fitness effects in other resources by competing them against their ancestor in five novel carbon sources. In these competitions, a relative fitness of one means that the genotype in question has a fitness equal to that of the ancestor, and thus there is no pleiotropy in such cases. The results of these competitions reveal a significant deviation from neutrality in all five novel resources for the mutants as a group (see confidence intervals, Fig. 1). In addition, the predominant fitness effect in these alternative environments is positive: in four of five resources, the grand mean fitness was significantly greater than one (see Fig. 1). Only in melibiose was fitness reduced on average, with a mean value of 0.92.

We also examined the fitness effects of the mutants individually, and we summarize these results in Table 2. Using a two-tailed t-test, we determined which mutants differed significantly in fitness from an ancestral value of 1.0 in each of the five novel resources. (For comparison, we also present the results for glucose.) The results confirm the findings based on the mutants as a group: where fitness effects differed

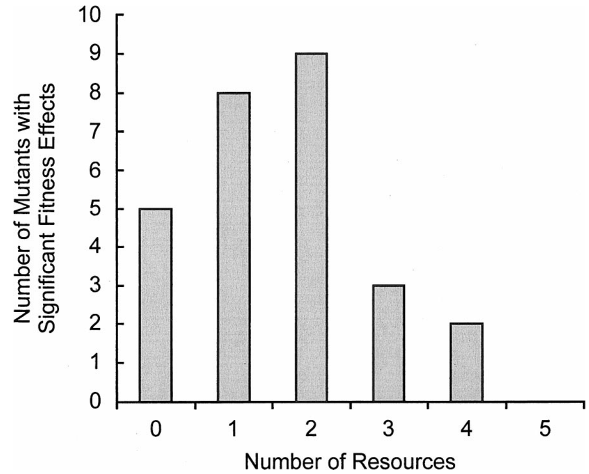

FIG. 2. Histogram showing the number of novel resources (of five) in which mutants had significant fitness effects.

significantly from the ancestor, the form of that pleiotropy was predominantly positive. Although 18 of 27 mutants had estimated fitness less than that of the ancestor in melibiose, only three of them were statistically significant. If we instead examine the extent of pleiotropy on a per mutant basis, we see that 22 of 27 mutants differed significantly from neutrality in at least one resource, and roughly half differed from the ancestor in two or more resources (Fig. 2). Thus, pleiotropy was common among these beneficial mutations, and its form was predominantly positive.

Using a two-way analysis of variance, we sought to address simultaneously the contributions of genotype and resource to the fitness effects seen in novel resources. The results revealed a highly significant genotype × resource interaction (Table 3, $P < 0.0001$). However, the data also indicated a significant departure from the assumption of homogeneity of variances (Levene's test, $F = 2.69$, df $= 133$, $P < 0.0001$). This effect was driven in large part by fitness estimates in melibiose, where the variances both within and among genotypes were much higher than in other resources. To determine whether melibiose was the source of the significant interaction, we re-ran the analysis, but this time excluded the melibiose data. The results show that the interaction was weaker, but still statistically significant ($F_{78,215} = 1.43$, $P = 0.0237$). The presence of a significant interaction between genotype and resource indicates that the mutants varied in their responses to different resources. The ANOVA also indicated significant main effects of resource and genotype (both $P < 0.0001$; Table 3), but the interpretation of main effects is problematic when the interaction is significant. To examine the main effects more closely, we performed sets of

TABLE 2. Categorization of the mutant fitness effects by resource. Mean relative fitness was assessed based on three independent estimates of fitness in each resource, except glucose, which was based on 15 estimates. Mutants were determined to be beneficial or deleterious if their fitness differed significantly from 1.0 based on a two-tailed t-test.

Criterion:	Glucose	NAG	Mannitol	Maltose	Galactose	Melibiose
Mean relative fitness ≥ 1	27	27	27	25	20	9
Number significantly beneficial	27	14	12	8	5	0
Number significantly deleterious	0	0	0	0	0	3

TABLE 3. Two-way analysis of variance for fitness of mutants in the five novel resources. Genotype and resource were treated as random and fixed effects, respectively.

Source	df	MS	F	P
Genotype	26	0.08427	6.46	<0.0001
Resource	4	1.68592	51.90	<0.0001
Genotype × resource	104	0.03250	2.49	<0.0001
Error	269	0.01305		

one-way ANOVAs to examine the effect of genotype within each resource, and the effect of resource for each genotype. Results are summarized in the Appendix, Tables A1 and A2, respectively. These analyses confirm that both genotype and resource were important. The effect of genotype was significant in four of the five novel resources, and the effect of resource was significant for 18 of the 27 mutants tested.

Do Pleiotropic Effects Scale with the Primary Effect?

To address whether the pleiotropic fitness effects scale with the magnitude of their direct effects, we calculated the correlation between fitness in glucose and each of the novel resources. For three resources (NAG, mannitol, and maltose), this correlation was highly significant and positive, even after correction for multiple comparisons (Fig. 3). That is, larger benefits in glucose were associated with larger benefits in these novel resources. For the two other resources, galactose and melibiose, no significant correlation was detected. The melibiose data were significantly nonnormal (Shapiro-Wilkes $W = 0.969$, $P = 0.003$), so we also calculated a nonparametric Spearman's rank correlation for those data. This test was also nonsignificant ($r_s = 0.092$, df = 25, $P = 0.650$). Finally, to assess the robustness of our results, we performed a bootstrap analysis by resampling with replacement from our 27 mutants and recalculating the various correlation coefficients 1000 times. With a resample size of 27 mutants, this method produced similar results to our parametric tests. The significant results remained so even when our resampled population was reduced to as few as 10 mutants, indicating that our results were robust even to small samples (data not shown).

It is interesting to note that the two resources that failed to show statistically significant correlations were the same resources in which average fitness values were lowest. One possibility for the failure to observe a significant correlation is that the mutants with deleterious fitness effects were obscuring a correlation among mutants with beneficial effects, and vice versa, such that positive and negative correlations in different subsets of the mutants were canceling each other out. To address this possibility, we divided the data into two groups: mutants with mean fitness greater than one and mutants with mean fitness less than one. We then calculated separate correlations for these two groups in melibiose and galactose. This analysis allows us to ask whether there is a general trend for mutations with positive pleiotropic effects to scale, and for those with negative effects to scale inversely, regardless of the statistical significance of the individual mutations.

In galactose, the correlation among mutants with fitness greater than one was positive and significant ($r = 0.459$, df $= 18$, $P = 0.042$), but the correlation for mutants with fitness less than one was also positive and not significant ($r = 0.590$, df $= 5$, $P = 0.164$). In melibiose, the pattern was similar: removing deleterious mutants resulted in a significant positive correlation ($r = 0.767$, df $= 7$, $P = 0.016$), but removing mutants with beneficial effects resulted in a correlation that was weakly positive and nonsignificant ($r = 0.111$, df $= 16$, $P = 0.662$). These results suggest that the deleterious effects of some mutants obscured a positive correlation among mutations with beneficial effects. However, the opposite pattern was not observed: that is, mutants with deleterious pleiotropic effects were not inversely correlated with their effects in glucose. Although the sample size of deleterious pleiotropic mutants was small, even the sign of correlation was not consistent with our hypothesis.

The positive correlations indicate the existence of a predictable, quantitative relationship between the magnitudes of a mutation's direct and pleiotropic effects in three resources. They do not, however, tell us how these effects scale—that is, whether pleiotropic effects tend to be smaller or larger than their corresponding primary effects. To address the issue of scaling, we calculated the slope of the regression of fitness in each novel resource on fitness in glucose (Table 4). Because our independent variable (fitness in glucose) was measured with error, we performed a Model II reduced major axis regression. The slope of this regression is mathematically equivalent to that of a simple linear regression, scaled by the correlation coefficient of the two variables. The analysis is therefore only appropriate for those comparisons for which there exists a statistically significant correlation—here, for NAG, mannitol, and maltose. Table 4 shows that all three slopes were similar in magnitude and did not differ significantly from a slope of one. Thus, the positive pleiotropic effects were usually similar in magnitude to their direct effects.

DISCUSSION

Micromutational models of adaptation rest on assumptions about the prevalence, magnitude, and form of pleiotropic effects associated with beneficial mutations, but there is little direct evidence on which to base these assumptions. Here we examine the prevalence, magnitude, and form of pleiotropic effects associated with a sample of spontaneous beneficial mutations in *E. coli*. Specifically, we ask whether the pleiotropic effects tend to be beneficial or deleterious, and whether they scale in proportion to the primary effects of the mutations. To do so, we examined a collection of mutants, each of which bears a mutation that confers a benefit in glucose. We determined the primary and pleiotropic effects of each mutation by measuring each mutant genotype's fitness in its selective glucose environment and five novel resource environments.

We find that pleiotropic effects of these mutations were common, with the majority having a significant fitness effect in two or more resources. The predominant form of this pleiotropy was positive. We did detect some antagonistic pleiotropy, but it was primarily limited to a single resource, melibiose, and the fitness effects were significant for only three of the 27 mutants tested. However, these antagonistic

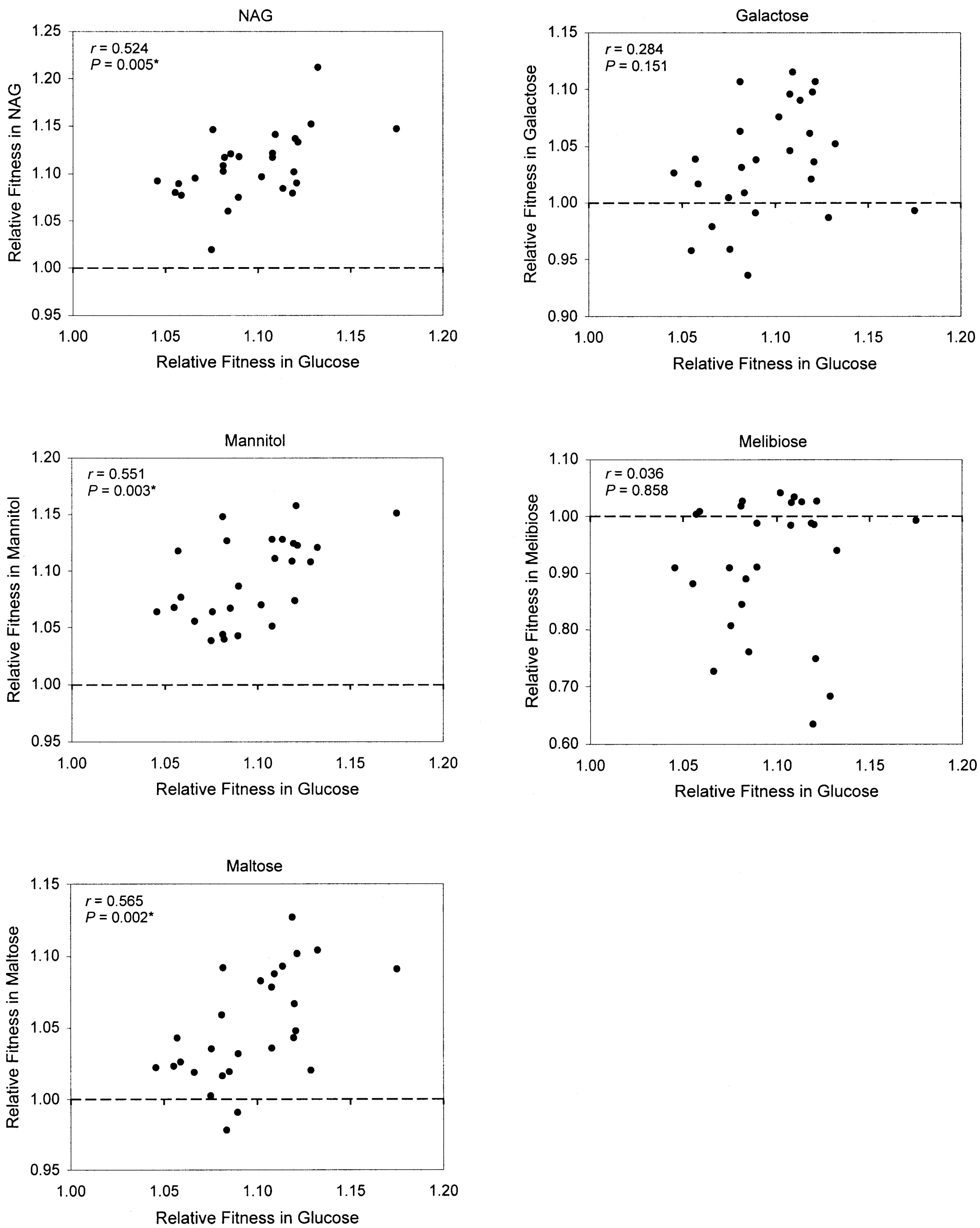

FIG. 3. Relationship between fitness in glucose and in five novel resources. Correlation coefficients and their statistical significance are shown on each graph. Asterisk indicates significance following sequential Bonferroni correction.

TABLE 4. Estimated model II regression coefficients (b_{RMA}) for fitness in novel resources versus fitness in glucose and their 95% confidence limits. A regression coefficient was calculated only if there was a significant underlying correlation.

		95% confidence limits	
Resource	b_{RMA}	Lower	Upper
NAG	1.247	0.809	1.685
Mannitol	1.278	0.838	1.717
Maltose	1.315	0.868	1.762

effects were at times very pronounced, with some mutants suffering fitness reductions of more than 30% relative to the ancestor in this resource.

In those resources where pleiotropic effects were positive, the pleiotropic effects of mutations were similar in magnitude to their direct effects, resulting in statistically significant correlations and slopes that did not differ significantly from one. By contrast, deleterious pleiotropic effects did not show a significant inverse correlation with the direct effect. However, the number of these mutations was small, which limited our statistical power. Individual deleterious pleiotropic effects—when they existed—were much larger than their direct effects; the three mutants that were significantly deleterious in melibiose had fitness detriments that were, on average, more than twice their corresponding benefits in glucose. These few examples provide some weak support for the hypothesis that deleterious pleiotropic effects will be disproportionate to their direct beneficial effect. Nevertheless, there were many large beneficial mutations that lacked deleterious pleiotropic effects in any of the resources tested.

One might suggest that the correlated responses reflect hitchhiking by mutations at other genomic sites rather than pleiotropic effects of the beneficial mutations per se. However, this possibility is unlikely for several reasons, two of which are most powerful. First, Lenski et al. (2003) sequenced 36 randomly chosen gene regions in bacteria sampled from all 12 of the long-term lines after 20,000 generations. From these data, they estimated a mutation rate for the ancestor of 1.44×10^{-10} per bp per cell generation. Given that the genome contains 4.64×10^{6} bp, this rate corresponds to a genomic rate of 6.68×10^{-4} mutations per cell generation. For the purpose of illustration, let us assume that (1) almost every mutation is selectively neutral in the glucose environment, and (2) the selective sweep whereby each beneficial mutant arose led to an effective population size of one for the duration. Even under these extreme assumptions, which maximize the potential for hitchhiking, the 400-generation experiments used to select the clones bearing beneficial mutations would have allowed a secondary mutation to hitchhike in only roughly a quarter ($\approx 400 \times 6.68 \times 10^{-4}$) of the cases. The real proportion was presumably much lower. Hence, it is unlikely that more than a handful of the clones carrying the beneficial mutations have any secondary mutations in their genomes. Second, the vast majority of random mutations—including these hypothetical hitchhikers—should be deleterious, not beneficial, when tested in environments that expose their phenotypic effects. However, the preponderance of correlated responses that we observed in the alternative resource environments were positive, not negative. These data are thus incompatible with the hypothesis that secondary, hitchhiking mutations generated the correlated responses. For these reasons, we are quite confident that the overall patterns of performance we measured on the alternative resources reflect pleiotropic effects of the beneficial mutations selected on glucose.

Our choice of resources was motivated by an earlier study of populations that evolved for 2000 generations on glucose, many of which showed reduced fitness on several of the resources used here (Travisano and Lenski 1996). In that study, five of 12 populations had reduced fitness on maltose, 11 of 12 had reduced fitness on galactose, and all 12 populations showed fitness reductions on melibiose. Our results show a similar qualitative pattern, in that fewer deleterious pleiotropic effects were detected in maltose than in galactose, and fewer in galactose than melibiose. However, our results fail to show that deleterious pleiotropy is common among beneficial mutations, at least with respect to resource use. Given this finding, there are two explanations for the observation that fitness declines were more common over the long term. One possibility is the mutations were not representative of those occurring over longer time periods. We can speculate that there may be differences between mutations that arise earlier versus later (with the latter possibly having smaller or more targeted effects), but there is no obvious reason to think that the distribution of deleterious pleiotropic effects would also differ. One possibility is that later mutations are more likely to be compensatory and, thus, might be more likely to have deleterious pleiotropic effects. However, mutations identified to date in the long-term populations appear to be beneficial even on the ancestral background, and therefore compensatory mutations are unlikely to explain our results (V. Cooper et al. 2001; T. Cooper et al. 2003; Crozat et al. 2005). A more parsimonious explanation is that our sample was representative of the possible mutations and, in fact, only a minority has deleterious pleiotropic effects on fitness in novel resources. This result implies that the parallel correlated declines in the long-term populations occurred simply because there is a higher probability of having sampled one or more of the subset of beneficial mutations with deleterious pleiotropic effects with increasing time. We might have expected that a similar relationship would hold for resources where fitness effects were primarily positive, that is, that only a small subset of mutations would be responsible for the positive correlated responses. Contrary to this expectation, however, pleiotropic effects were common in these resources and uniform in their direction.

One explanation for the predominance of positive pleiotropy is that the mutations confer increased fitness to aspects of the environment that are common to all resource regimes, for instance, incubator temperature, daily transfer cycles, and the like. Although these mutations may confer such general benefits, the significant interaction of genotype and resource—as well as the paucity of mutations that were beneficial in melibiose—indicates that at least some of these mutations have resource-specific effects. A second (and not mutually exclusive) possibility is that the mutations enhance competitive ability for glucose, but that glucose is functionally similar to some of the other resources. This possibility was also suggested by Travisano and Lenski (1996), who noted that similarity in performance on alternative resources following long-term evolution in glucose was highest for those resources that shared their mechanisms of uptake with those used for glucose. Their explanation is supported here by the fact that pleiotropy was most often positive for mannitol and NAG, which share with glucose the phosphotransferase system (PTS) for transport across the inner membrane of *E. coli*. Melibiose and galactose do not involve the PTS, which may explain why fitness effects were generally lower in these resources. However, the high fitness values observed in maltose are more puzzling in this regard. Maltose is not a PTS sugar and also uses a different porin from that of glucose to diffuse across the outer membrane. Moreover, in a long-term study involving multiple beneficial substitutions, Travisano and Lenski (1996) saw no systematic correlated gains in fitness on maltose, with some populations showing reduced fitness in maltose and others showing fitness increases. However, maltose is a glucose dimer, and once inside the cell and cleaved, its subsequent catabolism should be identical to that of glucose. Thus, while the pattern of pleiotropic effects seen in NAG and mannitol points to the PTS as a possible target of selection, the finding of positive pleiotropy in maltose suggests that at least some of the mutations included in this study may target later catabolic steps, or otherwise produce similar benefits in glucose and maltose (Travisano 1997). In general, we expect that the greater the similarity in the uptake and catabolism between the novel resources and glucose, the greater the probability that mutations will have similar fitness effects in the two resources.

Despite differences in the fitness effects of mutations in PTS versus non-PTS resources, there remains substantial heterogeneity in the correlated responses, even when we limit our consideration to non-PTS resources only. For example, galactose and melibiose are both non-PTS resources, and yet beneficial mutations were, on average, positive in galactose and deleterious in melibiose. This result is inconsistent with any single model of how these mutations affect fitness. Travisano and Lenski (1996) reached a similar conclusion, but we note that two additional explanations that could account for the variability in their experiment—drift and epistatic interactions among beneficial mutations—are not in force here. Thus, our study shows that variability in the correlated responses derives solely from the direct effects of different beneficial mutations within the same genetic background.

We emphasize that there are several important caveats to our findings. First, although our results show that positive pleiotropic effects were more common than deleterious pleiotropic effects, our assessment is clearly affected by the choice of resources: it is easy to imagine that a different set of environmental conditions might have produced different results. Moreover, by examining the fitness effects of these mutations on novel resources, we limited the scope of pleiotropy to a set of functionally related traits. In fact, it seems likely that these mutations have effects on other unmeasured traits, and thus may entail fitness consequences in other environments, including natural ones (Service and Rose 1985; Fry 2003). Pleiotropy can also arise from trade-offs among fitness components within a given resource environment, and this would be an interesting area for further study. A logical set of candidates for this sort of pleiotropy would be performance levels during the distinct series of physiological stages associated with the feast and famine conditions of our serial transfer regime. Following transfer to fresh medium, there is a period of acclimatization and preparation for growth (lag phase), followed by a period of maximal growth (exponential phase), decelerating growth, and, finally, survival after resources have been depleted (stationary phase). Hypothetically, for example, a beneficial mutation might shorten the duration of lag time, thereby speeding the transition to exponential growth, but carry the pleiotropic cost of reduced ability to survive during stationary phase. However, a previous study using populations evolved for 2000 generations under the same conditions found that these populations were generally improved for several fitness components, with little evidence for reduced performance during any other phase (Vasi et al. 1994).

Finally, an important finding of the current study is the variability in the responses of these mutants to different resources. This observation suggests that the chance occurrence of different mutations in different populations could be an important determinant of their future evolutionary directions (Mani and Clarke 1990; Travisano and Lenski 1996). Moreover, the variation among mutations in the form and extent of their pleiotropy raises interesting questions about the mechanistic bases of these effects and, in turn, about the genetic bases of these adaptations. The wealth of knowledge regarding the physiology and genetics of *E. coli* makes it an ideal system to begin establishing a clear mechanistic link between the action of natural selection on different phenotypes and the underlying genetic changes. Accordingly, we have begun sequencing several candidate genes in each of these mutants. One of these candidates is *spoT*, in which Cooper et al. (2003) found beneficial mutations in eight of 12 populations that evolved for 20,000 generations under the same conditions and from the same ancestor as the current study. In a follow-up to this paper, we hope to address the molecular basis of these adaptive changes and the extent to which differences in the pattern of their pleiotropic effects may reflect different underlying genetic changes.

Acknowledgments

We thank T. Cooper, D. Misevic, A. Orr, R. Woods, and two anonymous reviewers for helpful comments and feedback on an earlier version of this manuscript. We also thank N. Hajela for valuable technical assistance. This work was supported by a National Science Foundation grant (DEB-9981397) to REL.

Literature Cited

Barton, N. H., and P. D. Keightley. 2002. Understanding quantitative genetic variation. Nat. Rev. Genet. 3:11–21.
Cooper, T. F., D. E. Rozen, and R. E. Lenski. 2003. Parallel changes in gene expression after 20,000 generations of evolution in *Escherichia coli*. Proc. Natl. Acad. Sci. USA 100:1072–1077.
Cooper, V. S., and R. E. Lenski. 2000. The population genetics of ecological specialization in evolving *Escherichia coli* populations. Nature 407:736–739.
Cooper, V. S., D. Schneider, M. Blot, and R. E. Lenski. 2001. Mechanisms causing rapid and parallel losses of ribose catabolism in evolving populations of *Escherichia coli* B. J. Bacteriol. 183:2834–2841.
Crozat, E., N. Philippe, R. E. Lenski, J. Geiselmann, and D. Schneider. 2005. Long-term experimental evolution in *Escherichia coli*. XII. DNA topology as a key target of selection. Genetics 169: 523–532.
Elena, S. F., V. S. Cooper, and R. E. Lenski. 1996. Punctuated evolution caused by selection of rare beneficial mutations. Science 272:1802–1804.
Fisher, R. A. 1958. The genetical theory of natural selection. Dover, New York.
Fry, J. D. 2003. Detecting ecological trade-offs using selection experiments. Ecology 84:1672–1678.
Futuyma, D. J., and G. Moreno. 1988. The evolution of ecological specialization. Annu. Rev. Ecol. Syst. 19:207–233.
Gerrish, P. J., and R. E. Lenski. 1998. The fate of competing beneficial mutations in an asexual population. Genetica 103: 127–144.
Jaenike, J. 1990. Host specialization in phytophagous insects. Annu. Rev. Ecol. Syst. 21:243–273.
Lande, R. 1983. The response to selection on major and minor mutations affecting a metrical trait. Heredity 50:47–65.
Lenski, R. E., M. R. Rose, S. C. Simpson, and S. C. Tadler. 1991. Long-term experimental evolution in *Escherichia coli*. I. Adaptation and divergence during 2,000 generations. Am. Nat. 138: 1315–1341.
Lenski, R. E., C. L. Winkworth, and M. A. Riley. 2003. Rates of DNA sequence evolution in experimental populations of *Escherichia coli* during 20,000 generations. J. Mol. Evol. 56:498–508.

Mani, G. S., and B. C. Clarke. 1990. Mutational order: a major stochastic process in evolution. Proc. R. Soc. Lond. B 240: 29–37.

Orr, H. A., and J. A. Coyne. 1992. The genetics of adaptation: a reassessment. Am. Nat. 140:725–742.

Otto, S. P. 2004. Two steps forward, one step back: the pleiotropic effects of favored alleles. Proc. R. Soc. Lond. B 271:705–714.

Postma, P. W., J. W. Lengeler, and G. R. Jacobson. 1996. Phosphoenolpyruvate:carbohydrate phosphotransferase systems. Pp. 1149–1173 *in* F. C. Neidhardt, ed. *Escherichia coli* and *Salmonella*: cellular and molecular biology. ASM Press, Washington, DC.

Remold, S. K., and R. E. Lenski. 2001. Contribution of individual random mutations to genotype-by-environment interactions in *Escherichia coli*. Proc. Natl. Acad. Sci. USA 98:11388–11393.

Rice, W. R. 1989. Analyzing tables of statistical tests. Evolution 43:223–225.

Rose, M. R. 1991. Evolutionary biology of aging. Oxford Univ. Press, Oxford, U.K.

Rozen, D. E., J. A. G. M. de Visser, and P. J. Gerrish. 2002. Fitness effects of fixed beneficial mutations in microbial populations. Curr. Biol. 12:1040–1045.

SAS Institute. 1999. SAS Ver. 8. SAS Institute, Inc., Cary, NC.

Scheffé, H. 1959. The analysis of variance. Wiley, New York.

Service, P. M., and M. R. Rose. 1985. Genetic covariation among life-history components: the effect of novel environments. Evolution 39:943–945.

Sokal, R. R., and F. J. Rohlf. 1995. Biometry. Freeman, New York.

SYSTAT. 2002. SYSTAT Ver. 10.2. SYSTAT Software, Inc., Richmond, CA.

Travisano, M. 1997. Long-term experimental evolution in *Escherichia coli*. V. Environmental constraints on adaptation and divergence. Genetics 146:471–479.

Travisano, M., and R. E. Lenski. 1996. Long-term experimental evolution in *Escherichia coli*. IV. Targets of selection and the specificity of adaptation. Genetics 143:15–26.

Travisano, M., F. Vasi, and R. E. Lenski. 1995. Long-term experimental evolution in *Escherichia coli*. III. Variation among replicate populations in correlated responses to novel environments. Evolution 49:189–200.

Vasi, F., M. Travisano, and R. E. Lenski. 1994. Long-term experimental evolution in *Escherichia coli*. II. Changes in life-history traits during adaptation to a seasonal environment. Am. Nat. 144:432–456.

Wingreen, N. S., J. Miller, and E. C. Cox. 2003. Scaling of mutational effects in models for pleiotropy. Genetics 164: 1221–1228.

Corresponding Editor: S. Elena

Appendix

Table A1. Results of one-way ANOVAs testing the effect of genotype in each novel resource.

Resource	df	MS (Genotype)	MS (Error)	F	P
NAG	26, 54	0.0040	0.0027	1.49	0.1098
Mannitol	26, 54	0.0042	0.0022	1.88	0.0258
Maltose	26, 54	0.0044	0.0021	2.08	0.0119
Galactose	26, 53	0.0072	0.0021	3.46	<0.0001
Melibiose	26, 54	0.0427	0.0078	5.47	<0.0001

Table A2. Results of one-way ANOVAs for the fitness effect of each genotype across resources.

Genotype	df	MS (Resource)	MS (Error)	F	P
9962	4, 9	0.0097	0.0013	7.38	0.0064
9968	4, 10	0.0041	0.0023	1.80	0.2064
9970	4, 10	0.0092	0.0019	4.84	0.0197
9972	4, 10	0.0491	0.0058	8.44	0.0030
9974	4, 10	0.0300	0.0040	7.44	0.0048
9976	4, 10	0.0053	0.0041	1.29	0.3363
9978	4, 10	0.0033	0.0040	0.83	0.5369
9980	4, 10	0.0013	0.0020	0.63	0.6497
9982	4, 10	0.1013	0.0062	16.25	0.0002
9984	4, 10	0.0049	0.0057	0.87	0.5142
9986	4, 10	0.0048	0.0018	2.61	0.0997
9988	4, 10	0.0033	0.0022	1.47	0.2821
9990	4, 10	0.0185	0.0042	4.47	0.0250
9992	4, 10	0.0205	0.0023	9.02	0.0024
9994	4, 10	0.0736	0.0030	24.24	<0.0001
9996	4, 10	0.0087	0.0020	4.39	0.0262
9998	4, 10	0.0592	0.0104	5.70	0.0118
10000	4, 10	0.0077	0.0016	4.95	0.0183
10002	4, 10	0.0044	0.0025	1.74	0.2185
10004	4, 10	0.0075	0.0016	4.59	0.0232
10006	4, 10	0.1205	0.0040	30.01	<0.0001
10008	4, 10	0.0060	0.0020	3.08	0.0677
10012	4, 10	0.0145	0.0036	4.07	0.0328
10014	4, 10	0.0118	0.0020	5.79	0.0112
10016	4, 10	0.0438	0.0034	12.85	0.0006
10018	4, 10	0.0632	0.0056	11.23	0.0010
10020	4, 10	0.0238	0.0018	13.05	0.0006